

Sang Yoon Lee

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PERSONAL DATA

- ✓ Birth: 19th February, 1997, in Republic of Korea
- ✓ Nationality: Korean
- ✓ Language: Korean (native), English (fluent)
- ✓ Military Service: Honorably discharged as Sergeant (December 2018)

EDUCATION

Sep. 2025

University of Bristol (MSc Robotics)

Bristol, United Kingdom

- Relevant Coursework: Machine Vision, Advanced Control & Dynamics, Robot Learning (DMP/GMM/GMR), AI for Robotics, Robotic Science & Systems, Robotics Fundamentals, Human-Robot Interaction, Assistive Robotics
- Master's Thesis: Spatialized Audio Neurofeedback-based Calibration of a Motor Imagery EEG-BCI for Robot Control (in progress)
- Current Average : 75.3% (Distinction, dissertation in progress)

Mar. 2016 ~ Feb. 2022

Kyunghee University

Seoul, Korea

1st Bachelor of Electronic Engineering
GPA 3.39/4.5

RESEARCH EXPERIENCES

Master's Thesis Research

Spatialized Audio Neurofeedback-based Calibration of a Motor Imagery EEG-BCI for Robot Control

- Investigating alternative neurofeedback modalities to traditional visual feedback for motor imagery BCI calibration, reducing cognitive load and attentional resource competition
- Designing and implementing a 5-phase experimental protocol with 30 BCI-naive participants comparing discrete vs. continuous spatialized audio feedback paradigms against visual control
- Developing an 8-channel EEG signal processing pipeline (OpenBCI Cyton, 250 Hz) focused on C3/C4/Cz sensorimotor sites for left/right-hand motor imagery classification
- Validating neurofeedback effectiveness through progressive testing: volitional control assessment, simulated robotic arm pick-and-place tasks, and physical 1-DOF robotic arm control
- Hypothesizing that spatialized audio feedback will match or exceed visual feedback accuracy with lower cognitive workload, while discrete feedback will yield more robust transfer to robotic control applications

Mobile Robot Leader-Follower Coordination (Robotic Science & Systems)

- Built a communication-free decentralized leader-follower system on two differential-drive robots (Pololu 3pi+, embedded C++): IR-based relative localization with cascaded SMA/EMA signal filtering and dual proportional distance/heading loops for formation control.

- Designed an IMU-triggered role-switching FSM that swaps leader/follower on collision, achieving >95% success across 16–22 cm dead-end widths (single-leader baseline degraded 100% → 0%) and 27.1% shorter travel distance.

Advanced Control & Dynamics: Sliding-Mode + LQR Hybrid (2-Link Manipulator)

- Derived the Lagrangian dynamics of a 2-link planar manipulator and designed a hybrid controller pairing boundary-layer (chattering-free) sliding-mode control with an LQR linearised about the operating point, on a 3-level switch rig comparing SMC+LQR / SMC / FLC / pure LQR.
- Reached steady-state RMSE of 1.20 / 2.26 mrad (J1/J2) under bounded disturbance — 8–36% lower than standalone SMC and $\sim 18\times$ tighter than pure LQR — with $\sim 41\%$ lower torque total-variation. Extended the design to the underactuated case via collocated partial feedback linearisation (PFL).

Robot Learning: Learning from Demonstration (DMP & GMR)

- Implemented Dynamic Movement Primitives (DMP) for trajectory learning with goal and temporal scaling, selecting the kernel count by sweep ($K=9$) in MATLAB.
- Encoded demonstrations with GMM/GMR (EM, fixed seed) and benchmarked against locally-weighted least squares: GMR cut reproduction RMSE by 66.6% (0.3745 vs 1.1224). Closed the loop through analytical inverse kinematics and a model-based PD (computed-torque) controller ($\omega_n=10$ rad/s, $\zeta=1$), tracking joint trajectories to $1.36e-2$ / $3.49e-3$ rad RMSE (J1/J2).

AI for Robotics: HeroBot MazeDungeon (three ML paradigms)

- Supervised: compared a 3-block CNN (BatchNorm + global average pooling) with KNN-on-HOG for 80×80 sprite classification over 5 entity classes, using McNemar's test and expected calibration error (ECE) for a rigorous comparison.
- Unsupervised & RL: clustered entities with K-Means vs. agglomerative hierarchical (gap statistic + stability) and trained Q-Learning vs. SARSA on an 8×8 maze with potential-based reward shaping (Ng et al., 1999); RANDOM_SEED=42 throughout.

Machine Vision: Orchard Apple Detection & Counting

Built a staged detector — CLAHE → HSV/Laplacian segmentation → overlapping tiling → YOLOv8 → non-max suppression - and ran a 5-version ablation on a 77-apple, IoU-matched evaluation set. Final pipeline reached F1 0.89 / precision 86.6% / recall 92.2%, cutting false positives from 54 to 11 boxes versus the raw-YOLO baseline (F1 0.67).

WORK EXPERIENCES

Worked as a PMIC Application Engineer at LX Semicon (May. 2022~ Aug. 2025)

- Developed Python-based automation programs for measurement instruments (oscilloscopes, DMMs, power supplies) using GPIB communication, reducing efficiency measurement evaluation time
- Conducted comprehensive PMIC characteristic analysis including BUCK/BOOST efficiency measurements, Line/Load regulation evaluation, and IP current consumption analysis
- Designed and produced custom PCB boards for PMIC testing and competitor evaluation using OrCad for circuit design and PCB artwork/routing
- Performed system-level analysis of PMIC power sequences and LS CLK characteristics on SOC, including defect analysis and RMA response using automated data processing

RESEARCH INTEREST

Brain-Computer Interfaces & Neurotechnology

- Motor imagery EEG signal processing, neurofeedback paradigms, BCI-based robotic control

Control Theory & Applications

- Optimal control, robust control, intelligent control, multi-sensor fusion

Robotics & Mechatronics

- Humanoid robots, mobile robots, multi-robot coordination, human-robot interaction

Computer Vision & Machine Learning

- Image processing using machine learning, visual perception for robotics

TECHNICAL STRENGTHS

Programming Languages

- C/C++, C#, Python, MATLAB/Simulink, Verilog (HDL)

Robotics & Control

- Machine Vision, EEG signal processing, BCI systems, robotic arm control, DMP/GMM, ROS2

Hardware & Electronics

- PCB design and artwork (OrCad), soldering, MCU interface systems (Atmega 128, MSP430 , STM32 Nucleo)

Measurement & Instrumentation

- GPIB automation, oscilloscopes, DMMs, power supply control, I2C protocol

AWARDS AND HONORS

- CREATIVE COMPREHENSIVE DESIGN COMPETITION AWARD, KYUNGHEE UNIVERSITY (2021)

LEADERSHIP & ACTIVITIES

- President, Micro Robot Academic Club, Kyunghee University (2021)
- Vice President, College of Electronics and Information, Kyunghee University (2022)
- Coding Instructor, Ondream School Davinci Class for elementary students (2022)
- D-Buddy Supporter, Daejeon Foreign Resident Center (2024)

REFEREES

Professor Giacinto Barresi (MSc Dissertation Supervisor)

- Professor, College of Arts, Technology and Environment School of Engineering
- Bristol Robotics Laboratory, University of Bristol
- Email: Giacinto.Barresi@uwe.ac.uk
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Professor Dong-han Kim (Bachelor Dissertation Supervisor)

- Professor, Department of Electronics Engineering
- College of Electronics and Information, Kyung Hee University
- Email: dhkim@khu.ac.kr